

How and Why AI Systems Fail

WHY THIS MODULE MATTERS

AI systems do not usually fail loudly. They fail by working exactly as designed on the wrong objective, or by quietly drifting away from the world they were trained on. This module covers the recognised patterns and how to spot them early.

4.1 Failure Is Normal, Not Exceptional

The useful mental model is not "will this system fail?" but "how will this system fail, who will notice, and how long will it take?" Every deployed model is wrong some of the time; governance is about the wrongness being visible, bounded and owned.

Documented AI failures rarely involve the system breaking. Far more often the system performs precisely as built while the organisation discovers, in public, that what it built was not what it needed. The three root causes below account for most of the published cases.

Data failures

The training data misrepresents the world the system will operate in. It may encode past discrimination, as in the Amazon case. It may under-represent a group, so accuracy is much lower for them. It may be historically accurate but describe a world that has since changed. The model is faithful to its data; the data is the problem.

Objective failures

The system optimises a measurable proxy for the thing you actually wanted. You wanted good hires and optimised for resemblance to past hires. You wanted patient need and optimised for past healthcare spending. You wanted engagement and optimised for time on page, and got outrage. The proxy is always easier to measure than the goal, which is why this failure is so persistent.

Deployment failures

The system is used differently from how it was validated. A tool built to prioritise a reviewer's queue becomes, in practice, the decision itself. A model validated on one population is rolled out to another. A pilot's careful human checks are dropped when the tool scales. Nothing about the model changed; its role did.

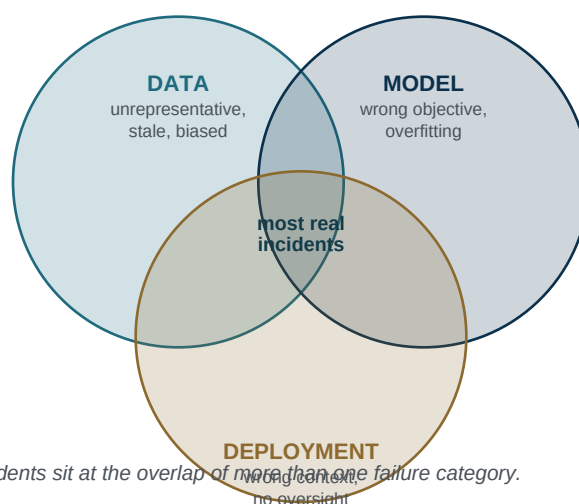


Figure 4.1 — Most documented incidents sit at the overlap of more than one failure category.

4.2 Drift: The Failure That Arrives Later

A model is a snapshot of the relationships present in its training data. The world moves on. Customer behaviour changes, populations shift, a competitor changes the market, a pandemic rewrites every pattern in the data at once. The model does not know this has happened and will continue producing confident outputs based on a world that no longer exists.

Drift is dangerous precisely because it is undramatic. Performance degrades gradually, often without any error, and the degradation is only visible if someone is measuring against real outcomes rather than against the model's own confidence. A system with no monitoring is not a stable system; it is an unobserved one.

CASE STUDY — THE DUTCH CHILDCARE BENEFITS SCANDAL

From 2013, the Dutch tax authority used an automated risk-scoring system to detect childcare benefit fraud. The system flagged families for investigation, and being flagged triggered aggressive recovery action — families were required to repay tens of thousands of euros, often for minor or non-existent administrative errors.

Dual nationality and low income were among the factors that raised a family's risk score. The result was that ethnic minority families were disproportionately targeted. Because the scoring was opaque, families could not find out why they had been flagged, and appeals took years.

More than twenty thousand families were affected. Some lost homes; over a thousand children were placed in state care. The Dutch government resigned in January 2021 over the scandal, and the data protection authority later issued substantial fines.

The governance failure was not a single bad decision but the absence of any mechanism to detect the pattern: no impact assessment on affected groups, no route for the affected to obtain an explanation, no escalation path when caseworkers noticed something was wrong, and no owner with the authority to halt the system.

4.3 Five Categories of Risk

Boards and risk committees tend to treat AI risk as a single line item, usually filed under technology. It is more useful to track five categories separately, because they have different owners, different early warning signs and different remedies.

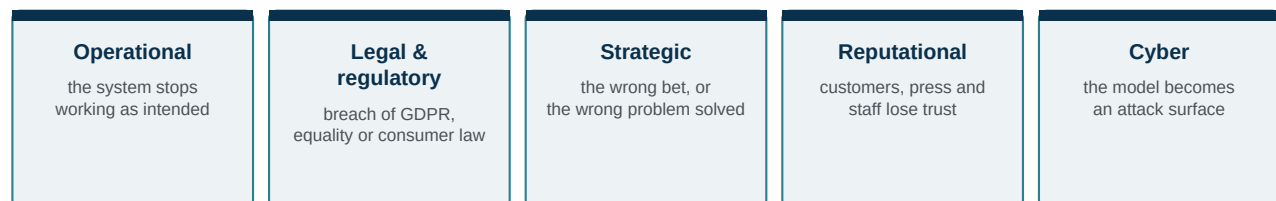


Figure 4.2 — Five categories of AI risk, best tracked separately.

- **Operational** — the system is inaccurate, unavailable or degrading, and business processes that depend on it fail with it.
- **Legal and regulatory** — the system processes personal data unlawfully, produces discriminatory outcomes, or breaches sector rules or consumer law.
- **Strategic** — the organisation becomes dependent on a capability, a vendor or a dataset it does not control and cannot readily replace.
- **Reputational** — the system produces an outcome that is defensible internally and indefensible publicly. This risk frequently crystallises fastest.
- **Security** — the model or its data becomes an attack surface, through data poisoning, model extraction, or prompt injection against generative systems.

4.4 Prioritising: Likelihood Against Impact

Not every AI system warrants the same scrutiny, and treating them all identically guarantees that the genuinely risky ones receive too little attention. The standard tool is a simple grid: how likely is this to go wrong, and how badly does it affect people when it does.

The important refinement for AI is that impact should be assessed as impact on the people subject to the system, not impact on the organisation. Those two often diverge sharply. A system that misclassifies a small number of benefit claims is low-impact to the organisation and potentially catastrophic to each affected household.

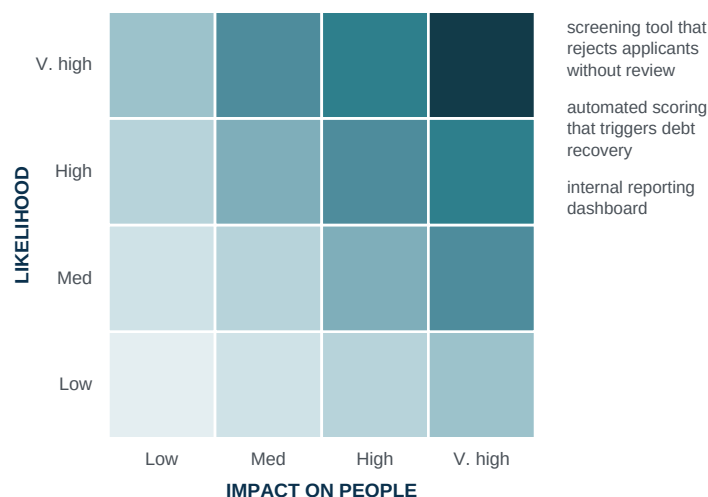


Figure 4.3 — Likelihood against impact on people, used to set review depth.

TRY THIS — PLOT THREE SYSTEMS

Take three AI systems in use around you and place each on the grid above, judging impact from the perspective of the person the system is applied to rather than the organisation running it.

Then ask, for the highest-placed one: what would be the first observable sign that it had started going wrong, and who would see that sign? If the answer is "a complaint", the system has no monitoring — it has an audience.

KEY TAKEAWAYS

- Most AI failures are the system working correctly on the wrong objective, not the system breaking.
- The three root causes are unrepresentative or historically biased data, optimising a proxy instead of the real goal, and using the system differently from how it was validated.
- Drift is degradation without error; it is invisible unless performance is measured against real-world outcomes.
- Track AI risk in five separate categories — operational, legal, strategic, reputational and security — because they have different owners and different warning signs.
- Assess impact from the perspective of the person subject to the decision, not the organisation making it.

FURTHER READING

- Dutch Data Protection Authority — findings on the childcare benefits (toeslagenaffaire) case
- NIST — AI Risk Management Framework (AI RMF 1.0)
- House of Commons Library — briefing on the 2020 exam grading algorithm

CHECK-IN QUIZ

1. A pricing model trained in 2023 continues to run through a period of rapid market change. It reports no errors, and its confidence scores remain high, while its predictions steadily worsen. This is:

This is:

- A. An objective failure
- B. Drift
- C. A deployment failure
- D. A security failure

2. A content platform wanted to increase user satisfaction and optimised for time spent on page. Engagement rose, and so did complaints about inflammatory content. This is:

- A. An objective failure
- B. A data failure
- C. Drift
- D. A privacy failure

CHECK-IN QUIZ

3. An organisation becomes wholly dependent on one vendor's model, with no ability to move to an alternative. Which risk category does this fall into?

- A. Operational
- B. Reputational
- C. Security
- D. Strategic

4. In the Dutch childcare benefits case, the mechanism whose absence most directly compounded the harm was:

- A. A backup of the scoring database
- B. Any route for affected families to obtain an explanation and challenge the score
- C. A contract with the software supplier
- D. An annual accuracy report to parliament

5. You ask how a live system would first show signs of failing, and the answer is "a customer would complain". This tells you that the system has:

- A. Adequate monitoring through customer feedback
- B. A well-designed escalation route
- C. No monitoring — only an audience
- D. Low inherent risk